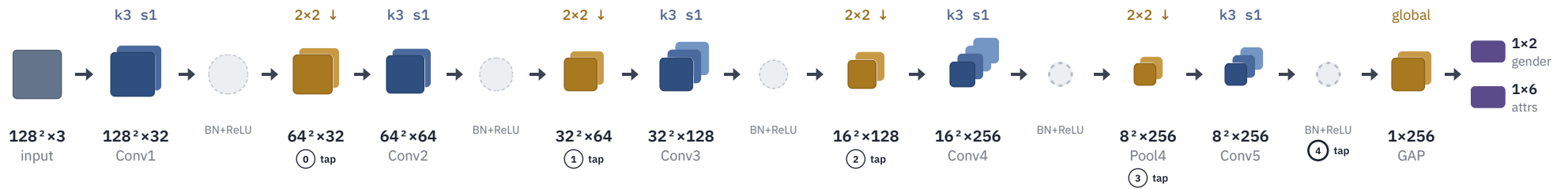
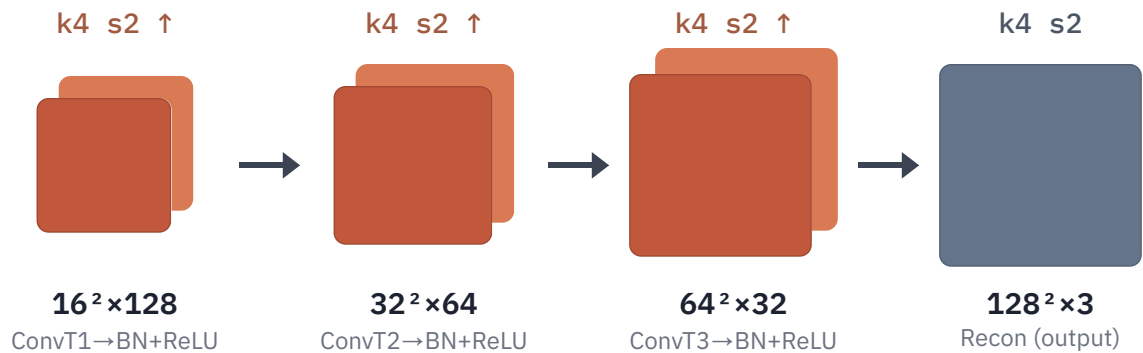


Shared CNN trunk ($128 \times 128 \times 3$ input) $\rightarrow$ classification heads + per-depth reconstruction taps



↓ tap 4 (bottleneck) — BlockDecoder, expanded example:



Taps 0–3 use frozen ConvT decoders with this same block structure, each independently reconstructing $128 \times 128 \times 3$ from its own depth.

TENSORS	Input / output	Conv feature map	BatchNorm + ReLU	Pool / GAP output	FC head
OPERATIONS	Data flow (Conv)	Max-Pool · downsample ↓	ConvTranspose · upsample ↑	Decoder tap (auxiliary)	

Box depth (Z) $\propto$ channels C · box W/H $\propto$ spatial extent